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Assessing system reliability in multi-grade production under different demand fulfilment policies via a multistate flow network

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ABSTRACT

The classification of products into grades based on the tolerances of key quality characteristics specified by customers is a common practice aligning with diverse customer needs and reducing scrap. This study is mainly to assess system reliability, the probability that the manufacturing system fulfils orders of all grades, as a performance index to determine the manageable order quantity to accept. The manufacturing system is first modelled as a multistate flow network (MFN) to analyse material flows across workstations required to satisfy the demands of all grades while incorporating the multiple capacities of each workstation caused by machine uncertainty. Then, two practical demand fulfilment policies are considered: satisfying demand for a grade with products of the same grade or through downward substitution. Lastly, the determination of the lower capacity vector (LCV) that meets the demands of all grades is utilised as an efficient method for calculating system reliability, with an algorithm proposed to generate the LCV under two policies. Several experiments are conducted to observe the changes in system reliability and profit of varying demands under the two policies, enabling managers of manufacturing companies to adjust order volumes and select the policy that best aligns with the objective of the company.

PRACTITIONER SUMMARY

A production system naturally produces products of different quality due to process variability. These products are often divided into different quality grades, each with its own demand. This study provides actionable insights to support production managers in making order management decisions when facing orders of products across these grades. Same-grade fulfilment (Policy I) and downward substitution (Policy II) are considered. System reliability, the probability that a manufacturing system can produce enough products to meet demand for all grades, is adopted as a performance index. By evaluating system reliability and associated profit, we observe a trade-off between the two policies: one performs better in profit, while the other performs better in system reliability. When profit is the priority, Policy I is preferred, although its system reliability is relatively low. For important or urgent orders, where unmet demand could hurt customer satisfaction or long-term relationships, Policy II is recommended to ensure higher system reliability. Managers should also be aware that prolonged storage under Policy I may gradually reduce its profit advantage. These findings help managers adjust order volumes and select the fulfilment policy that aligns with their objectives.

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Key quality characteristics; tolerances; downward substitution; multistate flow network (MFN); system reliability

1. Introduction

The same manufacturing system, operating under identical processes, may simultaneously produce products with varying quality levels (Kalay & Taşkın, 2021; Lu et al., 2019). Manufacturers typically aim to meet specified quality requirements, and products failing to conform are often discarded. However, customer expectations for the same product vary, with some accepting more relaxed standards (Gupta & Khanna, 2024; Kazemi et al., 2018). Ignoring these variations can result in significant losses from scrapped products and missed opportunities in lower-quality markets. To address this challenge, categorising products into multiple grades has been widely adopted as an effective approach to allocate products according to customer-specific quality requirements (Transchel et al., 2016). For instance, in the semiconductor industry, binning enables the sale of chips with slightly lower performance to customers with lower quality demands, thereby reducing scrap rates and enhancing overall profitability (Hsieh et al., 2011). Moreover, product grading creates demands for multiple quality levels, making it a critical concern for manufacturers to fulfil all grade-specific demands.

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When products are classified into multiple grades, demand fulfilment becomes more complex than in the single-grade case. Higher-quality products, due to tighter tolerances, are generally associated with superior functionality and higher market prices (Zhang et al., 2019). To maintain consistency between product value and customer expectations, orders are usually fulfilled with products of the same grade (Lu et al., 2019), referred to as same-grade fulfilment (named Policy I herein). However, the output of a particular grade may sometimes be insufficient to satisfy its demand, which can result in customer dissatisfaction and penalties. To address this issue, using higher-quality products to fulfil orders for lower-quality products has been adopted in practice (Ahiska et al., 2017; Wu et al., 2020), allowing manufacturers to meet demand while mitigating negative consequences, referred to downward substitution (named Policy II herein). The ability of a manufacturing system to fulfil multi-grade demand is directly influenced by these two policies. Therefore, their impacts on system performance are systematically assessed in this study.

A manufacturing system is a complex network in which raw materials and work-in-progress (WIP) are sequentially processed across multiple workstations to meet the required specifications. Whether the target quantity can be achieved depends on the capacity available at each workstation. When multiple grade orders are placed from diverse customers, order fulfilment capability becomes a critical concern for managers, since failing to satisfy demand may lead to breaches of contract, loss of customer trust, or even penalties (Craig et al., 2021; Shen et al., 2022). To assess such capability, many studies (Chang et al., 2024; Niu et al., 2023; Su et al., 2024; Yeh et al., 2023) have adopted system reliability as an effective metric, as it intuitively reflects the probability of successfully achieving the intended function of a system, while also allowing other considerations such as cost and time to be incorporated as constraints to address various managerial concerns. In this study, system reliability is defined as the probability that the manufacturing system can fulfil demand across all product grades, serving as a decision-support metric for managers in order acceptance.

In manufacturing systems, parallel configurations are commonplace, where each workstation is configured with multiple identical machines arranged in parallel (Ahmadi et al., 2024; Lin et al., 2016). Even if some machines are under maintenance, the workstation can continue to operate based on the number of machines in service, reflecting its multiple capacities. Therefore, the manufacturing system is modelled as a multistate flow network (MFN) to assess system reliability in many studies (Chang et al., 2017; Lin & Chang, 2012; Yang et al., 2024; Yao et al., 2024). By transforming the system into a network topology composed of nodes and edges, network analysis and the concept of minimal paths can be applied to analyse material flows that fulfil demand (Niu et al., 2020; Yeh, 2018). The concept of lower capacity vector (LCV), an efficient technique to calculate system reliability, is widely applied in related studies (Chang et al., 2024; Huang & Lin, 2024; Lin et al., 1995; Lin & Chang, 2012).

Although various MFN models have been developed to evaluate system reliability, existing studies primarily focus on single-product demands and have not addressed multi-grade demand and related fulfilment policies. Moreover, research on product grading and downward substitution has not considered the multistate nature of manufacturing systems. To address this gap, the aim of this study is to evaluate the system reliability of a manufacturing system under two demand fulfilment policies: same-grade fulfilment and downward substitution. When applying product grading, higher-quality grades naturally have higher value (market price), and demand is typically fulfilled using the same-grade products. However, due to variability in the manufacturing system, each grade is produced in a different proportion of the total output. Fulfilling demand using the same-grade products can result in either the overproduction of higher-quality grades or lower-quality grades, or in a shortfall of products for a specific grade. Compared with same-grade fulfilment, downward substitution is a widely adopted alternative in practice, provides greater flexibility in resource utilisation. When excess higher-grade products are available, they can be used to satisfy lower-grade demand. Thus, overproduction of a specific grade is less likely to occur under downward substitution; however, revenue loss will be incurred due to high-value products being sold at lower prices. Since the probability of successful order fulfilment and profit are regarded as major considerations by manufacturing managers, these two demand fulfilment policies are taken into account in this study. An algorithm is developed to generate the lower capacity vector (LCV) for either policy, enabling efficient assessment of system reliability. Profit analysis is further conducted for both policies to observe the influence on system reliability. By integrating system reliability and profit analysis, managerial insights are provided to assist in the selection of appropriate fulfilment policies and the adjustment of order quantities for effective order acceptance decisions.

The remainder of this study is organised as follows: In Section 2, the relevant works of literature are reviewed and analysed. The model formulation is represented in Section 3. The proposed algorithm

to generate the LCV under two practical policies for evaluating system reliability is demonstrated in Section 4. In Section 5, the system reliability of a manufacturing system producing ball screw pairs is assessed to validate the proposed algorithm, followed by experimental analysis and discussion. In Section 6, profit analysis and managerial implications are presented. Finally, conclusions are drawn in Section 7.

2. Literature review

The relevant literature for this study is focused on product grading, downward substitution, and system reliability assessment. The relevance of this study to these research areas is discussed below.

2.1. Product grading

Products are typically classified through full inspection based on multiple acceptable standards or tolerances for key quality characteristics (KQCs) or attributes to match them to their corresponding applications (Kolus & Duffuaa, 2025). Several studies (Kolus & Duffuaa, 2025; Lahmar et al., 2025; Park et al., 2021; Raza & Turiac, 2016; Tang, 1990; Zhang et al., 2019) have developed models to assess product grades and evaluate their impact on profit. Zhang et al. (2019) proposed a piecewise quadratic loss function for hierarchical products to estimate expected losses across different grades and select the most economical material. Lahmar et al. (2025) classified remanufactured products into different quality grades, each with distinct manufacturing processes and CO_2 emission coefficients, and applied the non-dominated genetic sorting algorithm (NSGA-II), entropy, and technique for order preference by similarity to ideal solution (TOPSIS) to prioritise high-quality order demand while minimising cost and emissions. The improvement of product grading efficiency has also attracted attention in several studies (Hsieh et al., 2011; Jia et al., 2021; Kim & Lee, 2023). Hsieh et al. (2011) proposed an adaptive error-tolerance-based testing method to enhance yield by grading chips according to error rates while minimising test costs. Jia et al. (2021) used 27-dimensional RGB colour-difference vectors as input and applied partial least squares discriminant analysis to build a "small-sample, high-dimensional" model that reduces dimensionality while enhancing classification performance, enabling quick identification of tea product categories and grades. Kim and Lee (2023) developed a neural-network-based approach using in-mold sensors and vibration data to predict the quality grade of each moulded part in real time, enabling timely reduction of defects and waste. While these studies investigate how to more efficiently classify products, improve grading efficiency, and achieve higher profitability, they either do not address demand satisfaction or assume fixed capacity values, without considering the multistate nature of capacity in manufacturing systems.

2.2. Downward substitution

Downward substitution is commonly applied in manufacturing and inventory management when orders for lower-quality products cannot be fulfilled and are instead satisfied using available higher-quality or premium products (Ahiska et al., 2017; Freeman et al., 2021; Luo & Chen, 2022; Transchel et al., 2016; Wu et al., 2020; Yu et al., 2017; Zheng & Gershwin, 2015). Downward substitution between products produced on distinct production lines has been the focus of some studies (Ahiska et al., 2017; Notz & Pibernik, 2022; Yu et al., 2017) supporting production planning. Notz and Pibernik (2022) developed a kERM-based two-stage capacity planning model with fixed capacities for multiple production lines, each producing a specific product grade; downward substitution is incorporated, allowing higher-level production lines to satisfy demands requiring lower expertise. Yu et al. (2017) used a Markov decision process to determine the optimal joint policy for a manufacturing system with premium and standard production lines, considering production and substitution decisions alongside dynamic pricing strategies. Some studies (Luo & Chen, 2022; Wu et al., 2020) have investigated sourcing products of different quality directly from suppliers to determine optimal ordering policies. Wu et al. (2020) developed a mathematical model that incorporates product substitution and dual sourcing strategies under partial production line disruptions, using a greedy allocation algorithm to determine optimal allocation and minimise expected costs. Luo and Chen (2022) investigated a single-period hospital platelet inventory problem, considering heterogeneous age-based demand, random supply of young platelets, and the effect of unidirectional substitution. They determined the optimal ordering policies using dynamic programming and analysed their impact on service level and expected cost. Overall, these studies have primarily addressed issues related to maximising profit in production planning and purchasing from the perspective of inventory management. In addition, studies related to manufacturing system capacity either ignore uncertainty and treat capacity as a fixed value, or consider only the capacity of an entire production line in binary states, without accounting for multiple possible capacities. In this study, downward substitution is treated as an

alternative policy within a single production line, under which only internally produced higher-quality products are used to satisfy shortfalls in lower-grade products. The ability of the manufacturing system to fulfil all grade orders under two policies is assessed considering multiple capacities and corresponding probabilities.

2.3. System reliability assessment

Over the past decades, the system reliability assessment has received increasing attention, as it offers a comprehensive overview of the performance of real-world systems. Because such systems often exhibit complex configurations and are concerned with fulfilling demand rather than simply maintaining operation, network analysis is commonly employed in system reliability assessment models to facilitate the effective representation and analysis of material or information flows (Lin et al., 2016; Wang et al., 2022). For system reliability assessment in manufacturing systems, many studies (Chang, 2019; Huang & Lin, 2024; Liao et al., 2023; Lin & Chang, 2012; Yang et al., 2024; Yao et al., 2024) model the practical manufacturing system as a MFN based on different production scenarios, with multiple capacities and their associated probabilities used to evaluate system reliability. Yao et al. (2024) proposed an extended MFN that integrates a Markov process to capture degradation driven by heterogeneous feedstock and thereby more accurately assess system reliability. Huang and Lin (2024) further developed a long short-term memory recurrent neural network model designed to enhance the efficiency of system reliability prediction and provide a basis for maintenance decision-making. As many studies assume buffer to be infinite, to address finite buffers which cause material flow blockage and starvation between workstations and buffers, Chang (2019) incorporated a Monte Carlo simulation to estimate the system reliability. However, only the ability to fulfil a single demand is considered in these studies, while scenarios involving multiple product grades have not been addressed. To bridge this gap, system reliability in fulfilling orders across multiple product grades is assessed using the proposed method, with the two different demand fulfilment policies considered specifically in the context of product grading.

3. Model formulation and system reliability assessment

In this section, the problem description and assumptions of this study are first outlined and discussed. Next, the approach of product grading is presented. Then, the formulation of a manufacturing system into an MFN is introduced. Lastly, the system reliability assessment method is presented.

3.1. Problem descriptions and assumptions

Different from the references (Ahiska et al., 2017; Notz & Pibernik, 2022; Yu et al., 2017), this study considers that products of different grades are produced on the same production line. Product grading differentiates products according to heterogeneous customer requirements. For example, in smartphone camera lens module production, modules are classified into two grades based on three KQCs: modulation transfer functions (MTFs) at the centre, middle, and edge field. MTF reflects lens resolution and is expressed as a percentage: 80%–100% for excellent grade, 50%–80% for acceptable grade. A module is classified as an excellent grade only if all field MTFs are 80%–100%; if any field MTF is 50%–80% and the others are 80%–100%, it is an acceptable grade; otherwise, it is defective. Excellent-grade modules are primarily intended for top-tier devices, whereas acceptable-grade modules are used for lower-tier devices. Customers demand both grades, with excellent-grade products featuring tighter tolerances and higher market value. Therefore, the problem addressed in this study is assessing system reliability for fulfilling multi-grade orders; in other words, it examines the extent that a manufacturing system has sufficient capacity to meet multi-grade demand under machine uncertainties and limited resources. We consider a manufacturing system that consists of multiple workstations arranged in series, where raw materials or WIPs must visit all workstations sequentially to meet the specified tolerances of all KQCs. Typically, each workstation is composed of one or multiple identical machines arranged in a parallel configuration. To facilitate analysis of material flow, the manufacturing system is modelled as a network topology, with workstations represented as edges and inspections/buffer storage represented as nodes. Then, the system reliability of the manufacturing system is assessed for multi-grade order acceptance decision. The problem is studied under the following assumptions:

1. The capacities of edges are statistically independent (Lin et al., 2016; Yang et al., 2022).
2. Non-destructive quality inspections, movement of WIP, and buffer storage between workstations will not cause conforming items to become defective.
3. The specifications of KQCs are statistically independent. (He et al., 2017; Wang et al., 2022; Yang et al., 2024).

4. Machine availability, yield rates, and percentage of products in each grade are calculated based on point estimation using historical data.

Assumption 1 means that the machines at different workstations are of different types; their capacities are inherently independent and do not influence each other. Assumption 2 indicates that WIPs are unlikely to be damaged during storage and transportation, as they are typically protected by fixtures or shipping trays. Assumption 3 states that in cases where different key quality characteristics (KQCs) are determined by separate processes or machines, they do not affect each other. For instance, the dimension and surface roughness of a part may result from different machining operations, making it reasonable to consider them independent. Assumption 4 is established because, although the estimates of the parameters mentioned above fall within an interval, point estimation is used to provide a simplified and representative value, facilitating model calculations and analysis.

Moreover, to prepare for the introduction of the LCV concept for system reliability assessment in Section 3.4, the following vector operation rules are first defined. Suppose there are two capacity vectors $Y = (y_1, y_2, \ldots, y_n)$ and $Z = (z_1, z_2, \ldots, z_n)$,

1. $Y \leq Z$ if and only if $y_i \leq z_i$ for each $i = 1, 2, \ldots, n$.
2. $Y < Z$ if and only if $Y \leq Z$, and there exists at least one i such that $y_i < z_i$.

Both capacity vectors Y and Z represent one possible capacity state of all workstations in the manufacturing system. For example, consider a manufacturing system with three workstations, where workstations 1, 2, and 3 contain three machines each, with per-machine capacities of 50, 55, and 60 respectively. Suppose $Y = (50, 55, 60)$ and $Z = (150, 165, 180)$. Y represents the state where only one machine is operating in each workstation, while Z represents the state where all three machines are operating in each workstation.

3.2. Product grading

The produced products are classified into grades based on KQCs. Generally, quality characteristics can be classified into three types: nominal-the-best (NTB), smaller-the-best (STB), and larger-the-best (LTB) characteristics. A product is accepted only if all specifications meet the defined tolerances, which are set either by both a lower specification limit (LSL) and an upper specification limit (USL), or by a target value together with either the LSL or the USL. Specifications outside these tolerances are considered defective. Some defective products may still be sold at a discount or accepted by other customers, depending on quality and price. Accordingly, products can be classified into multiple grades based on these tolerances. The simplest case of categorising products into grades is based on the tolerances of a single KQC. For grade g, the LSL and USL are expressed as LSL_g and USL_g, respectively, where $g = 1, 2, \ldots, h$. Products are deemed as grade g when the specification of the specific KQC that exceeds the tolerance of grade $g-1$ but remains within the gth tolerance. An instance of grading products based on each type of KQC is shown in Figure 1.

For more general cases of multiple KQCs, customers may specify various tolerances and have differing degrees of acceptance for imperfect products. To simplify production and product offerings, products can be categorised into several grades based on

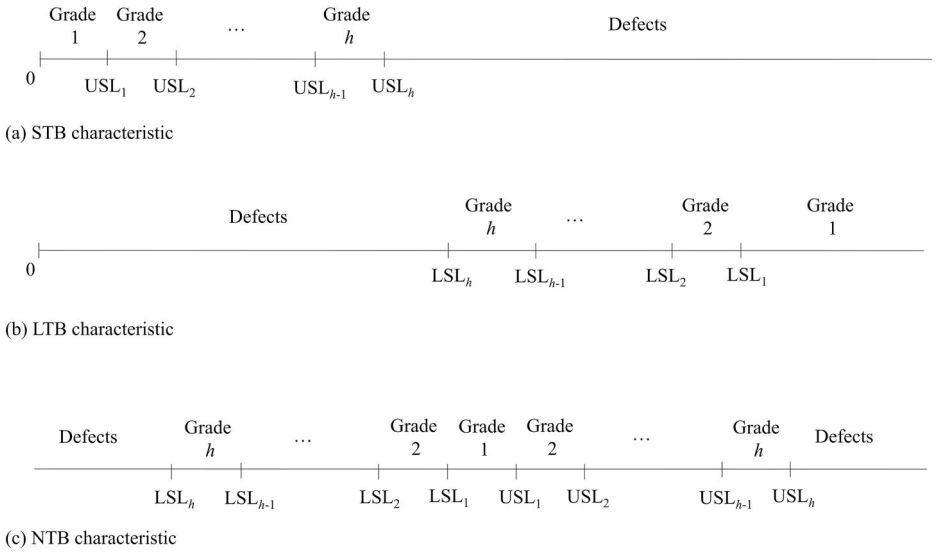


Figure 1. Grading products based on a single KQC.

tolerances specified by customers, ranging from strict to lenient. Products that meet the first tolerance of all KQCs are defined as grade 1. If the specification of any KQC of the product exceeds the first tolerances of all KQCs but still meets those of the second tolerances, these products are deemed as grade 2. This logic can be extended to cases with g tolerances of all KQCs that the specification of any KQC of a product exceeds the $(g-1)$th tolerances but still meets those of the gth tolerances is defined as grade g. Suppose each KQC has h tolerances and there are β KQCs in total, $\underline{s}_g^\alpha$ and $\overline{s}_g^\alpha$ are utilised to denote the minimal and maximal values of the gth tolerance for the αth KQC where $\alpha = 1, 2, \ldots, \beta$ and $g = 1, 2, \ldots, h$. Without loss of generality, let $\underline{s}_1^\alpha \geq \underline{s}_2^\alpha \geq \ldots \geq \underline{s}_h^\alpha$ and $\overline{s}_1^\alpha \leq \overline{s}_2^\alpha \leq \ldots \leq \overline{s}_h^\alpha$.

The percentage of products in each grade can be determined based on tolerances and historical specification data collected during previous manufacturing processes. The specifications of the αth KQC are represented as a random variable X_α with x_α being a single observed specification, where $\alpha = 1, 2, \ldots, \beta$; these random variables are generally assumed to follow a normal distribution (Raza & Turiac, 2016), denoted as $X_\alpha \sim N(\hat{\mu}_{X_\alpha}, \hat{\sigma}_{X_\alpha})$; the distribution can also be obtained from the historical data. The mean $\hat{\mu}_{X_\alpha}$ and standard deivation $\hat{\sigma}_{X_\alpha}$ are calculated with historical data of the αth KQC. The probability of the αth KQC meets the gth tolerance is calculated as below:

$$q_g^\alpha = \Phi\left(\frac{\overline{s}_g^\alpha - \hat{\mu}_{X_\alpha}}{\hat{\sigma}_{X_\alpha}}\right) - \Phi\left(\frac{\underline{s}_g^\alpha - \hat{\mu}_{X_\alpha}}{\hat{\sigma}_{X_\alpha}}\right), g = 1, 2, \ldots, h. \quad (1)$$

where $\Phi(\cdot)$ represents the cumulative distribution function of standard normal distribution. Moreover, based on Assumption 1 that KQCs are assumed to be statistically independent, the percentage of grade 1 products Q_1 can be calculated as follows:

$$Q_1 = \prod_{\alpha=1}^{\beta} q_1^\alpha, \ \alpha = 1, 2, \ldots, \beta. \quad (2)$$

Since grade 2 products are those with their specification falls within the second tolerances of all KQCs and at least one specification exceed its first tolerance, the percentage of grade 2 can be obtained by first calculating the probability that all KQCs' specification falls within the second tolerances of all KQCs then subtracting the percentage Q_1 of grade 1. Similarly, the percentage Q_g of grade g can be calculated as the percentage of products meeting the gth tolerances of all KQCs subtracting the cumulative percentage of from grade 1 to grade $g-1$ as below:

$$Q_g = \prod_{\alpha=1}^{\beta} q_g^\alpha - \prod_{\alpha=1}^{\beta} q_{g-1}^\alpha, \text{ where } g = 2, \ldots, h. \quad (3)$$

3.3. Formulation of an MFN and fulfilment policies

Let (**N, E, S**) denote a multistate flow network (MFN), where **N** denotes the set of nodes representing buffers and inspection procedures, $\mathbf{E} = \{e_i | i = 1, 2, \ldots, n\}$ with e_i denoting the ith edge (workstation), $\mathbf{S} = \{[\underline{s}_g^\alpha, \overline{s}_g^\alpha] | \alpha = 1, 2, \ldots, \beta, g = 1, 2, \ldots, h\}$ with $\underline{s}_g^\alpha$ and $\overline{s}_g^\alpha$ representing the minimum and maximum of the gth tolerance of the αth KQC, respectively. The capacity z_i of the ith ($i = 1, 2, \ldots, n$) workstation takes values from 0, v_i, $2v_i$, ⋯, ($u_i v_i = m_i$) where v_i is machine capacity, u_i is the maximal number of machines in operation. Thus, m_i is the maximal capacity of ith workstation. The demand of all h grades is expressed as $D = (d_1, d_2, \ldots, d_h)$.

When products are categorised into grades and sold at distinct prices, the demand D can be satisfied by one of the following two policies:

Policy I: The demand of each grade can be only fulfilled with the output of the same grade, as shown in Figure 2(a).

Policy II: The demand for any grade can be fulfilled with downward substitution, when there are superior grades available, as shown in Figure 2(b) (Gallego et al., 2006).

Policy I mandates that products of grade g be used exclusively to fulfil the demand for grade g, $g = 1, 2, \ldots, h$. It ensures the consistency of product value and quality, thus meeting customer expectations. In contrast, Policy II provides greater flexibility by prioritising excess higher-quality products for lower-quality demands, enhancing resource utilisation, and reducing production burdens. However, it may lead to potential losses by using higher-quality products to meet demand for lower-quality ones.

3.4. System reliability assessment

System reliability is defined as the probability that a manufacturing system can provide adequate capacity to process the workloads (raw materials or WIPs need to be processed) outputting sufficient products for meeting the demands for all grades in a given period of time. The capacity of a workstation is determined by the number of operational machines multiplied by the machine capacity in a specified period of time. Therefore, the probability of a workstation providing a specific capacity can be obtained by the binomial distribution (Chang, 2022).

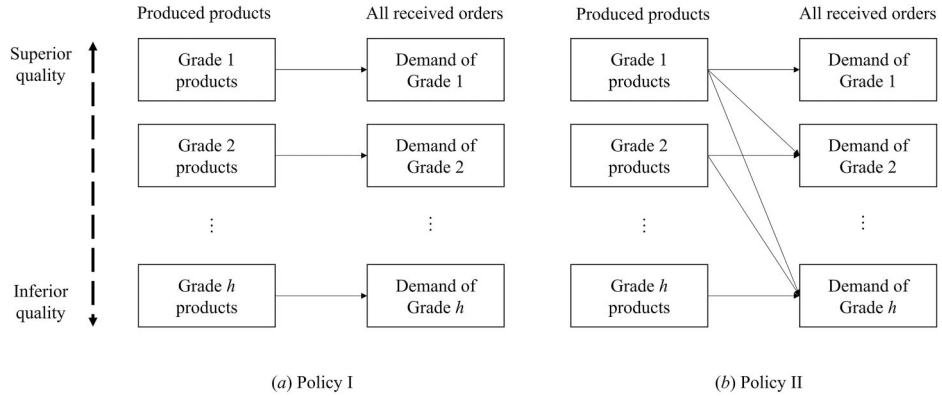


Figure 2. Demand fulfilment mechanisms under two policies.

Generally, the more input, the more likely the manufacturing system to produce enough amount of products to meet the demands as long as all workstations can collectively provide adequate capacity to process their corresponding workloads. Let θ_i represent the yield rate of the ith workstation, and let I denote the amount of raw material input at the first workstation. The workload of the first workstation, denoted as w_1, is equal to I. The workload for the second to the jth workstations, denoted as w_j, is equal to the output of the previous workstation, which is calculated as the amount of raw material input multiplied by cumulative yield rates of the preceding workstation, as shown below:

$$w_j = I \times \prod_{i=1}^{j-1} \theta_i, \; j = 2, 3, \ldots, n. \tag{4}$$

The workload of each workstation should not surpass its maximal capacity, as shown below:

$$w_i \leq m_i \text{ where } i = 2, 3, \ldots, n. \tag{5}$$

In an MFN, the capacity of all workstations is represented by the capacity vector $Z = (z_i | \; i = 1, 2, \ldots, n)$, where z_i is the (current) capacity of the ith workstation. The probability $\Pr\{Z\}$ that the MFN can provide the capacities specified by a capacity vector Z can be calculated as $\Pr\{z_1\} \times \Pr\{z_2\} \times \ldots \times \Pr\{z_n\}$ by Assumption 1. Then, all capacity vectors that meet the demand D are denoted as a set Ω. System reliability can be calculated directly by enumerating all capacity vectors in Ω, then summing up all the probabilities that the MFN can provide the capacities of all capacity vectors shown below:

$$R_D = \sum\nolimits_{Z \in \Omega} \Pr\{Z\}. \tag{6}$$

A more efficient method is to assess system reliability with the LCV. A capacity vector $\underline{Z} = (\underline{z}_i | \; i = 1, 2, \ldots, n)$ is the LCV if there is any capacity vector Y that the output of at least one grade cannot be satisfied such $Y < \underline{Z}$. Then, the system reliability R_D can be quickly calculated with the LCV as presented below:

$$R_D = \prod_{i=1}^{n} \Pr\{z_i \geq \underline{z}_i\}. \tag{7}$$

The LCV is obtained by finding the smallest number t_i of machines in operation of workstation i that provides the adequate capacity to process the workload w_i, such that $\underline{z}_i = t_i v_i \geq w_i > (t_i - 1)v_i$.

4. Generation of LCV

In this section, the inputs required to satisfy demand D are first presented for both policies. The definitions of the LCV under each policy are then introduced, followed by an algorithm for generating the LCV.

4.1. Determination of input under Policy I

The output of each grade determines whether the demands can be fulfilled. The output O can be calculated by the multiplication of the amount I of raw material input and yield rate θ_i of the ith workstation where $i = 1, 2, \ldots, n$, as $O = I \times \prod_{i=1}^{n} \theta_i$. The output O_g of grade g can be calculated as output times the percentage Q_g of grade g, as shown below:

$$O_g = I \times Q_g, \text{ where } g = 1, 2, \ldots, h. \tag{8}$$

To produce sufficient O_g for meeting the demand d_g of grade g, the amount I_g of raw material input required to meet the demand of grade g is obtained by setting $O_g = d_g$ in Equation (8) and is derived as follows:

$$I_g = \frac{d_g}{Q_g \cdot \prod_{i=1}^{n} \theta_i}, g = 1, 2, \ldots, h., \tag{9}$$

However, the amount of raw material input that can output an adequate amount of grade g meeting

the demand d_g of grade g may not meet the demands for other grades due to the percentage of each grade. The amount of input I_g can only guarantee the corresponding output meets the demand of grade g. To ensure that the demand D is fully satisfied, the required amount I of raw material input must take maximum among all I_g ($g = 1, 2, \ldots, h$) is shown below:

$$I = \max\{I_1, I_2, \ldots, I_h\} = \max\left\{\frac{d_1}{Q_1 \cdot \prod_{i=1}^{n}\theta_i}, \frac{d_2}{Q_2 \cdot \prod_{i=1}^{n}\theta_i}, \ldots, \frac{d_h}{Q_h \cdot \prod_{i=1}^{n}\theta_i}\right\}. \quad (10)$$

4.2. Determination of input under Policy II

Downward substitution can be adopted when lower-quality products are insufficient to meet demand, but higher-quality products are available in excess. The excess amount and shortages of grade g is (O_g-d_g) where $g = 1, 2, \ldots, h$. When (O_g-d_g) > 0, it indicates an excess of products in that grade; when O_g-$d_g < 0$, it indicates that the demand for that grade has not yet been met. Even under Policy II, the demand d_1 of grade 1 cannot be met by other grades as grade 1 represents the highest value and quality. The input satisfying d_1 is obtained as below:

$$I_1 = \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}. \quad (11)$$

To calculate the amount of required input to fulfil the demands of all grades under Policy II, which allows downward substitution, we start with the simplest case, where products are classified into only two grades. The excess grade 1 products and the output of grade 2 products can be used to fulfil the demand of grade 2, namely $(O_1\text{-}d_1) + O_2 \geq d_2$. The input I'_2 satisfying the demand of grade 2 through downward substitution can be determined as below:

$$I'_2 = \frac{d_1 + d_2}{(Q_1 + Q_2) \times \prod_{i=1}^{n}\theta_i}. \quad (12)$$

It seems that the input I'_2 will satisfy the demand of two grades. However, when the input I equals I'_2, the corresponding output is not guaranteed to satisfy the demand d_1 of grade 1. To illustrate this phenomenon, Table 1 shows the data of amount of input and output for two scenarios of a manufacturing system, each with different percentages and the same demands for two grades. In scenario 1, the output of grade 1 is less than its demand when the input I equals I'_2, whereas the demand of grade 1 is met in scenario 2.

Thus, the input for guaranteeing both d_1 and d_2 must be the maximum of I_1 and $I_2{}'$, as shown below:

$$I = \max(I_1, I'_2) = \max\left(\frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}, \frac{d_1 + d_2}{(Q_1 + Q_2) \times \prod_{i=1}^{n}\theta_i}\right). \quad (13)$$

To validate Equation (13), Propositions 1, 2, and Lemma 1 are provided to further prove its correctness.

Proposition 1. If $\frac{d_2}{Q_2 \times \prod_{i=1}^{n}\theta_i} > \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}$, then $\frac{d_1+d_2}{(Q_1+Q_2) \times \prod_{i=1}^{n}\theta_i} > \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}$.

Proof: see in Appendix A.

Proposition 2. If $\frac{d_2}{Q_2 \times \prod_{i=1}^{n}\theta_i} < \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}$, then $\frac{d_1+d_2}{(Q_1+Q_2) \times \prod_{i=1}^{n}\theta_i} > \frac{d_2}{Q_2 \times \prod_{i=1}^{n}\theta_i}$.

Proof: see in Appendix B.

Lemma 1.

$$\min\left\{\frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}, \frac{d_2}{Q_2 \times \prod_{i=1}^{n}\theta_i}\right\} \leq \frac{d_1 + d_2}{(Q_1+Q_2) \times \prod_{i=1}^{n}\theta_i} \leq \max\left\{\frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i}, \frac{d_2}{Q_2 \times \prod_{i=1}^{n}\theta_i}\right\}$$

Table 1. Results of two scenarios for demand D = (80, 60) and $\prod_{i=1}^{n}\theta_i = 0.8$.

Scenario	Q_1	Q_2	Amount of raw material input I	Amount of grade 1 output O_1	Amount of grade 2 output O_2
1	0.2	0.4	$I_1 = \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i} = 500$	80	160
			$I'_2 = \frac{d_1+d_2}{(Q_1+Q_2) \times \prod_{i=1}^{n}\theta_i} = 291.6667$	46.6667	93.3333
2	0.4	0.2	$I_1 = \frac{d_1}{Q_1 \times \prod_{i=1}^{n}\theta_i} = 250$	80	40
			$I'_2 = \frac{d_1+d_2}{(Q_1+Q_2) \times \prod_{i=1}^{n}\theta_i} = 291.6667$	93.3333	46.6667

Proof: By Propositions 1 and 2, we conclude the proof.

Propositions 1 and 2 describe how the three input levels I_1, I_2 and I'_2 are related when $I_2 > I_1$ and $I_1 > I_2$, respectively. Lemma 1 integrates these two cases and provides an overall view of how I'_2 compares with I_1 and I_2. When downward substitution is allowed, the role of I_2 becomes less critical because I'_2 already reflects the input requirement for satisfying grade 2 demand under downward substitution. Based on Lemma 1 and the downward substitution mechanism that any shortage in grade 2 can be compensated by reallocating higher-quality products, the required material input is determined by taking the maximum of I_1 and I'_2.

According to the above derivation process, we can extend the results to the more general case where products are classified into h grades. Any grade ζ with excess products, namely O_ζ-$d_\zeta > 0$, can be used to meet the shortages of any grade η with $\zeta < \eta$. The fulfilment of demands of grade η can be represented as $\sum_{\zeta=1}^{\eta-1}(O_\zeta - d_\zeta) + O_\eta \geq d_\eta$, $1 \leq \zeta < \eta \leq h$, which is equivalent to $(O_1 + O_2 + \ldots + O_\eta) \geq (d_1 + d_2 + \ldots + d_\eta)$ and can be rewritten as follows:

$$I \times \left(\sum_{g=1}^{\eta} Q_g\right) \times \prod_{i=1}^{n} \theta_i \geq \sum_{g=1}^{\delta} d_g. \tag{14}$$

The input I'_η satisfies the demand of grade η through downward substitution can be determined as below:

$$I'\eta = \frac{\sum_{g=1}^{\eta} d_\eta}{\sum_{g=1}^{\eta} Q_g \cdot \prod_{i=1}^{n} \theta_i}, \text{ where } \eta = 2, 3, \ldots, h. \tag{15}$$

Based on the logic of classifying products into two grades, when the input is equal to I'_η, it only guarantees the demand of grade η. The input required to meet the demand of all grades is calculated similarly, by taking the maximum among I_1, $I'_2, \ldots, I'_h$, as shown below:

$$\begin{aligned} I = & \max(I_1, I'_2, \ldots, I'_h) \\ = & \max\left(\frac{d_1}{Q_1 \times \prod_{i=1}^{n} \theta_i}, \frac{\sum_{g=1}^{2} d_q}{\left(\sum_{g=1}^{2} Q_g\right) \times \prod_{i=1}^{n} \theta_i}, \ldots, \frac{\sum_{g=1}^{h} d_q}{\left(\sum_{g=1}^{h} Q_g\right) \times \prod_{i=1}^{n} \theta_i}\right). \end{aligned} \tag{16}$$

4.3. Definition of LCV

The LCV represents the lowest vector in Ω. However, the capacity vectors satisfying the demand D may vary depending on the policy applied. For clarity, let Ω_{I} be the set of all feasible capacity vectors Z satisfying the demand D under Policy I, that is $\Omega_I = \left\{Z \middle| O_g(Z) \geq d_g,\ g = 1, 2, \ldots, h\right\}$ where $O_g(Z)$ is the maximal output of grade g product under capacity vector Z. Similarly, let Ω_{II} be the set of all feasible capacity vectors Z satisfying the demand D under Policy II, $\Omega_{\mathrm{II}} = \left\{Z \middle| \sum_{g=1}^{\eta} O_g(Z) \geq \sum_{g=1}^{\eta} d_g,\ \eta = 1, 2, \ldots, h\right\}$. The definition of the LCV under two policies are defined as below,

Definition 1. LCV under Policy I

A capacity vector $\underline{Z}$ is considered as the LCV in Ω_{I} if and only if (i) $O_g(\underline{Z}) \geq d_g$ and (ii) $O_g(Y) < d_g$ for at least one g for any capacity vector Y such that $Y < \underline{Z}$.

Definition 2. LCV under Policy II

A capacity vector $\underline{Z} = (\underline{z}_1, \underline{z}_2, \ldots, \underline{z}_n)$ is an LCV in Ω_{II} if and only if (i) $\sum_{g=1}^{\eta} O_g(\underline{Z}) \geq \sum_{g=1}^{\eta} d_g$ and (ii) $\sum_{g=1}^{\eta} O_g(Y) < \sum_{g=1}^{\eta} d_g$ for at least one η and for any capacity vector Y such that $Y < \underline{Z}$.

4.4. Algorithm to generate LCV

The following algorithm is developed for generating the LCV under either Policy I or Policy II, where the shortages of a grade are replenished from available grades with superior quality or performance. The applied policy is denoted by ψ, where $\psi \in \{1, 2\}$, representing Policy I and Policy II, respectively. Below is the pseudocode of the proposed algorithm.

Algorithm: Generating LCV under either policy

Input: (N, E, S), $D = (d_1, d_2, \ldots, d_h)$, ψ
For $\alpha \leftarrow 1$ to β // Step 1. Calculate the percentage of the αth KQC meeting all tolerances, respectively.
  For $g \leftarrow 1$ to h
    $q_g^{\alpha} \leftarrow \Phi\left(\frac{\bar{s}_g^{\alpha} - \hat{\mu}_{X_\alpha}}{\hat{\sigma}_{X_\alpha}}\right) - \Phi\left(\frac{\underline{s}_g^{\alpha} - \hat{\mu}_{X_\alpha}}{\hat{\sigma}_{X_\alpha}}\right)$
  End For
End For
For $g \leftarrow 1$ to h // Step 2. Calculate the percentage of each grade.
  $Q_g \leftarrow \prod_{\alpha=1}^{\beta} q_g^{\alpha} - \prod_{\alpha=1}^{\beta} q_{g-1}^{\alpha}$
End For
If $\psi = 1$ //Step 3. Calculate the required amount of raw material meeting demands under policy ψ.
  $I \leftarrow \max\left(\frac{d_1}{Q_1 \cdot \prod_{i=1}^{n} \theta_i}, \frac{d_2}{Q_2 \cdot \prod_{i=1}^{n} \theta_i}, \ldots, \frac{d_h}{Q_h \cdot \prod_{i=1}^{n} \theta_i}\right)$
  $w_1 \leftarrow I$
ELSE
  $I \leftarrow \max\left(\frac{d_1}{Q_1 \times \prod_{i=1}^{n} \theta_i}, \frac{\sum_{g=1}^{h} d_g}{(\sum_{g=1}^{2} Q_g) \times \prod_{i=1}^{n} \theta_i}, \ldots, \frac{\sum_{g=1}^{h} d_g}{(\sum_{g=1}^{h} Q_g) \times \prod_{i=1}^{n} \theta_i}\right)$
  $w_1 \leftarrow I$
End IF
For $j \leftarrow 2$ to n// Step4. Determine the workload of all workstations.
  $w_j \leftarrow w_{j-1} \times \theta_{j-1}$
End For
IF $\forall i \in \{1, \ldots, n\}$, $w_i \leq m_i$ // Step 5. Check whether any workstation violates the maximal capacity constraint.
  For $i \leftarrow 1$ to n // Step 6. Determine the smallest number ti of machines in operation at workstation i that provides sufficient capacity to process the workload wi.
    For $t_i \leftarrow 1$ to u_i
      IF $t_i \times v_i \geq w_i > (t_i - 1) \times v_i$.
        $\underline{z}_i \leftarrow t_i \times v_i$
      End IF
    End For
  End For
Else $R_D \leftarrow 0$
End IF
Output: $\underline{Z} = (\underline{z}_1, \underline{z}_2, \ldots, \underline{z}_n)$ or $R_D = 0$

5. Application examples

In this section, the manufacturing system of the ball screw pair is utilised for validating the applicability of the proposed algorithm. System reliabilities under two policies are calculated with identical demands of two grades for comparing the difference between the two policies. Lastly, numerical experiments and discussions are conducted.

5.1. Background

The ball screw pair is a crucial component used in products across various fields, including machine tools, robotics, and measurement equipment manufacturing, as it provides precise and durable linear motion. This motion is achieved by converting the rotational movement of balls rolling along a helical groove into linear motion. The specifications of KQCs impact the functionality of ball screw pairs, influencing their precision, accuracy, and repeatability of movement. The five processes involved in forming all required quality characteristics and KQCs, namely left-end centre hole fine drilling (e_1), bottom diameter fine grinding (e_2), centre hole grinding (e_3), external diameter fine grinding (e_4), and screw diameter fine grinding (e_5), are illustrated in the formulated MFN, as shown in Figure 3. The multiple capacities and corresponding probability that each workstation can provide a specific capacity is shown in Table 2.

The tolerances and specification data of all KQCs of ball screw pairs are presented in Table 3. Based on these tolerances, ball screw pairs are classified into two grades, offering customers a selection according to their preferences. Although grade 2 is considered to be of a lower quality than grade 1, its reduced price makes it an acceptable option for some customers. Therefore, the manufacturing company may receive orders for both grades from diverse customers, and the demands for the two grades are obtained by summing all orders that are planned to be accepted.

5.2. System reliability assessment and experimental findings

Suppose the demands of the two grades are 550 and 350, the system reliability is 0.6036 under Policy I and 0.9486 under Policy II. The detailed calculations are provided in Appendices D and E, respectively. The system reliability under Policy II is approximately 34% higher than that under Policy I. To further understand the impacts of different demands under two policies, the required raw material, system reliability, and the excess/shortage of each grade are summarised in Tables 4 and 5, respectively. The system reliability under Policy II is higher than or equal

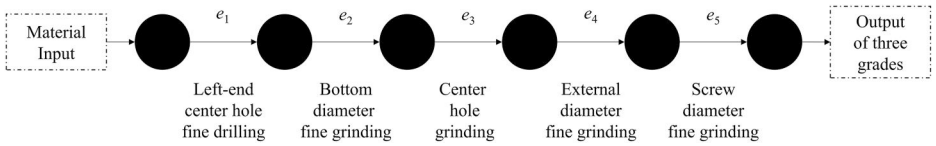


Figure 3. Ball screw manufacturing system.

Table 2. Probability table of workstations in ball screw pair manufacturing system.

Workstation	Machine capacity	Yield rate	Machine reliability	Workstation capacity	Probability
e_1	300	0.95	0.91	0	0.0000
				300	0.0000
				600	0.0001
				900	0.0017
				1200	0.0175
				1500	0.1061
				1800	0.3578
				2100	0.5168
e_2	350	0.97	0.92	0	0.0000
				350	0.0000
				700	0.0005
				1050	0.0080
				1400	0.0688
				1750	0.3164
				2100	0.6064
e_3	250	0.94	0.91	0	0.0000
				250	0.0000
				500	0.0001
				750	0.0017
				1000	0.0175
				1250	0.1061
				1500	0.3578
				1750	0.5168
e_4	225	0.96	0.93	0	0.0000
				225	0.0000
				450	0.0000
				675	0.0001
				900	0.0013
				1125	0.0134
				1350	0.0888
				1575	0.3370
				1800	0.5596
e_5	300	0.95	0.94	0	0.0000
				300	0.0000
				600	0.0002
				900	0.0036
				1200	0.0422
				1500	0.2642
				1800	0.6899

Table 3. Performance and tolerances of each KQC.

KQC		Mean and standard deviation $(\bar{\mu}_{X_\alpha}, \bar{\sigma}_{X_\alpha})$	The first tolerance $[\bar{s}_1^\alpha, \underline{s}_1^\alpha]$	The second tolerance $[\bar{s}_2^\alpha, \underline{s}_2^\alpha]$
$\alpha = 1$	Surface roughness	(0.0007, 0.0001)	[0, 0.0008]	[0, 0.0009]
$\alpha = 2$	Centre hole diameter	(36.124, 0.052)	[36, 36.2]	[35.98, 36.22]
$\alpha = 3$	External diameter	(26.86, 0.04)	[26.8, 27.05]	[26.78, 27.06]
$\alpha = 4$	Screw diameter	(40.481, 0.020)	[40.45, 40.51]	[40.44, 40.52]

to that under Policy I. Specifically, when the system reliability under the two policies is identical, such as $(d_1, d_2) = (650,250)$ and (700, 200), it indicates that no excess products are available for downward substitution. As a result, Policy II becomes essentially equivalent to Policy I under these demand scenarios. Excess products arise only under Policy I. In contrast, under Policy II, the amount of excess products may be zero when shortages of grade 2 are exactly offset by excess grade 1 through downward substitution, such as (d_1, d_2) equalling (500, 400) or (550, 350).

Differing demands D for the two grades may lead to different input values while resulting in the same system reliability. Motivated by this observation, we further evaluate the system reliability corresponding to all possible input levels, which reveals that when the input derived from demand D falls within a specific range, the system reliability remains unchanged, as illustrated in Figure 4. For instance, system reliability equals 0.6036 when the amount I of raw material input is in the range of [1559, 1627].

Policy II implements downward substitution whenever higher-quality products are available.

Table 4. System reliability and relevant data with varying demands under Policy I.

$D = (d_1, d_2)$	I	R_D	O_1-d_1	O_2-d_2
(400, 500)	2307.3693	0	739.4459	0
(450, 450)	2076.6323	0	575.5013	0
(500, 400)	1845.8954	0.0625	411.5567	0
(500, 300)	1384.4504	0.8346	183.6675	0
(400, 400)	1845.9339	0.0625	511.5567	0
(550, 350)	1615.1585	0.6036	247.6121	0
(600, 300)	1384.4216	0.8346	83.6675	0
(650, 250)	1316.2451	0.9359	0	35.2264
(700, 200)	1417.4947	0.8346	0	107.1669
(750, 150)	1518.7443	0.6633	0	179.1073
(750, 337)	1555.1669	0.6633	17.9865	0
(750, 338)	1559.7816	0.6036	20.2654	0
(750, 339)	1564.396	0.6036	22.5443	0
□	□	□	□	□
…	…	…	…	…
(750, 352)	1624.3880	0.6036	52.1699	0
(750, 353)	1629.0027	0.3566	54.4488	0

Note: The rows from $(d_1, d_2) = (750, d_2)$, where $d_2 = 340, 341, \ldots, 351$ are omitted for brevity, as the system reliability remains constant across all values of d_2.

Table 5. System reliability and relevant data with varying demands under Policy II.

$D = (d_1, d_2)$	I	R_D	O_1-d_1	O_2-d_2
(400, 500)	1266.6672	0.9486	225.5170	−225.5170
(450, 450)	1266.6672	0.9486	175.5170	−175.5170
(500, 400)	1266.6672	0.9486	125.5170	−125.5170
(500, 300)	1125.9264	0.9655	56.0151	−56.0151
(400, 400)	1125.9264	0.9655	156.0151	−156.0151
(550, 350)	1266.6672	0.9486	75.5170	−75.5170
(600, 300)	1266.6672	0.9486	25.5170	−25.5170
(650, 250)	1316.2451	0.9359	0	35.2264
(700, 200)	1417.4947	0.8346	0	107.1669
(750, 150)	1518.7443	0.6633	0	179.1073
(750, 337)	1529.8525	0.6633	5.4855	−5.48555
(750, 338)	1531.2599	0.6633	6.1806	−6.18056
(750, 339)	1532.6673	0.6633	6.8756	−6.87558
□	□	□	□	□
…	…	…	…	…
(750, 352)	1550.9637	0.6633	15.9108	−15.9108
(750, 353)	1552.3711	0.6633	16.6058	−16.6058

Note: The negative value of $(O_2$-$d_2)$ indicates the amount of products that are required to be replenished by excess products of grade 1 to meet the demand for grade 2 products; the rows from $(d_1, d_2) = (750, d_2)$, where $d_2 = 340, 341, \ldots, 351$ are omitted for brevity, as the system reliability remains constant across all values of d_2.

Although the effect of limited substitution—where only a portion of higher-quality products is used—has not been directly addressed in this study, its impact on system reliability can be obtained from the raw material inputs required under Policy I and Policy II. The required amount of raw material under Policy I is generally greater than that under Policy II. Any raw material input below the amount required under Policy I necessitates downward substitution; otherwise, the demand cannot be satisfied. In other words, varying the raw material input within the range between the amounts required under Policy I and Policy II represents different extents of limited downward substitution. Since the system reliability of inputting different amounts of raw material for production has been demonstrated in Figure 4, the variation in system reliability under different levels of limited downward substitution can therefore be observed. For example, according to Tables 4 and 5, given $D = (550, 350)$, the required amount of raw material input under Policy I is 1615.1585 and 1266.6672 under Policy II. Depending on the extent of limited downward substitution, system reliability can range from approximately 0.6 to 0.9.

Additionally, the above results pertain only to the simplest case, where products are categorised into two grades. Further experiments for the general case involve ball screw pairs of three grades (see Tables 6 and 7). For simplicity, the percentages of the three grades are assumed to be 0.5, 0.25, and 0.2, respectively. The results of the general case are largely consistent with those of the simplest case. The key difference is that, in the simplest case, any available excess products are fully used to offset shortages after substitution. In the general case, however, excess products may remain after substitution, as illustrated by $(d_1, d_2, d_3) = (400, 100, 200)$ or $(200, 400, 100)$.

6. Profit analysis and managerial implications

In this section, a profit analysis of the two policies is conducted. Based on the profit performance and system reliability results, several managerial implications are discussed to support decision-making.

6.1. Profit analysis

Regardless of whether demand is fulfilled using Policy I or Policy II, profit is always a primary concern for manufacturing managers. Therefore, the key cost components involved in the production process, including material input cost and processing cost are further analysed in this section. Compared with Policy II, Policy I is more likely to produce a larger number of excess products. Therefore, to ensure a fair comparison of profit between the two policies, both the revenue obtained from selling excess products and the inventory cost incurred before these products are sold are taken into account. Accordingly, under ideal conditions, excess products of grade g are assumed to be sold after t_g days. For clarity, the cost components considered in the profit analysis are summarised as follows:

(1) Material cost:

The amount of raw material input determines the material cost, which is calculated as $I \times MC$, where MC is the per-unit cost of a raw material.

(2) Manufacturing cost:

Manufacturing cost is driven by the amount of material or work-in-progress being handled at each

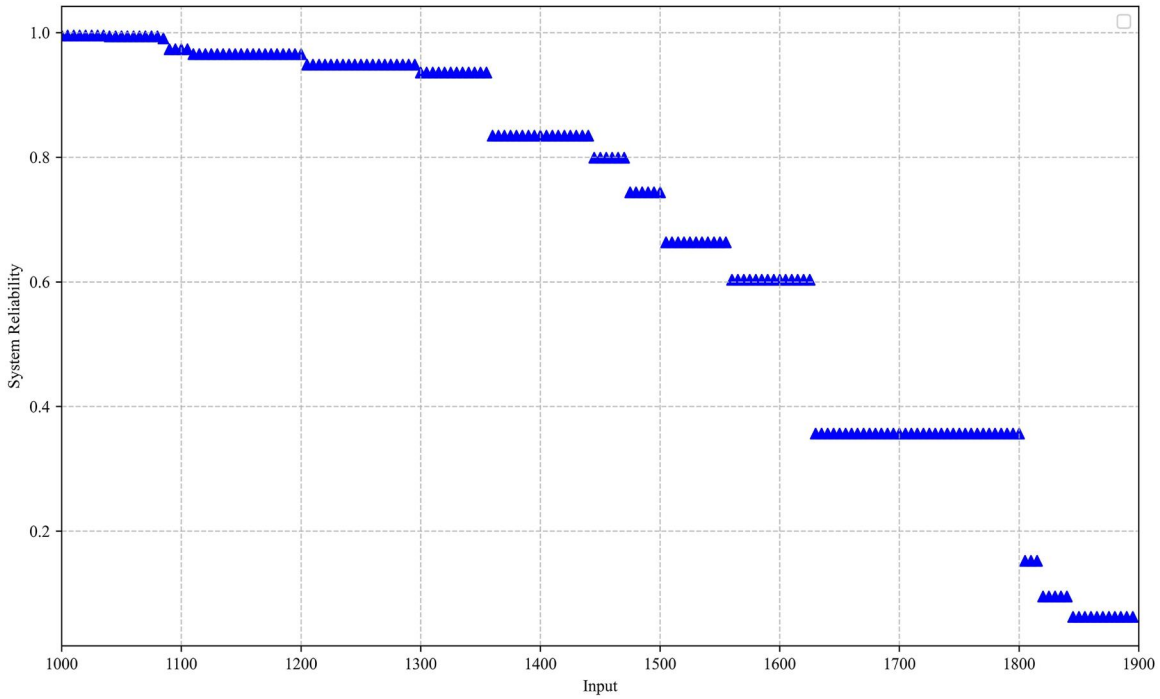


Figure 4. The relationship between the amount of input and system reliability.

Table 6. Experiments for the three-grade case under Policy I at different demands.

$D = (d_1, d_2, d_3)$	I	R_D	O_1-d_1	O_2-d_2	O_3-d_3
(250, 150, 100)	759.4937	0.9990	50	0	19.9
(150, 250, 100)	1265.8228	0.9487	350	0	100
(100, 250, 150)	1265.8228	0.9487	400	0	50
(250, 100, 150)	949.3671	0.9955	125	87.5	0
(150, 100, 250)	1582.2785	0.6036	475	212.5	0
(100, 150, 250)	1582.2785	0.6036	525	162.5	0
(400, 200, 100)	1012.6582	0.9955	0	0	60
(400, 100, 200)	1265.8228	0.9487	100	150	0
(200, 100, 400)	2531.6456	0	800	400	0
(200, 400, 100)	2025.3165	0	600	0	220
(100, 200, 400)	2531.6456	0	900	300	0
(100, 400, 200)	2025.3165	0	700	0	120

Table 7. Experiments for the three-grade case under Policy II at different demands.

$D = (d_1, d_2, d_3)$	I	R_D	O_1-d_1	O_2-d_2	O_3-d_3
(250, 150, 100)	675.1055	0.9990	16.6667	−16.6667	6.6667
(150, 250, 100)	675.1055	0.9997	116.6667	−116.6667	6.6667
(100, 250, 150)	666.2225	0.9997	163.1579	−118.4211	−44.7368
(250, 100, 150)	666.2225	0.9997	13.1579	31.5789	−44.7368
(150, 100, 250)	666.2225	0.9997	113.1579	31.5789	−144.7368
(100, 150, 250)	666.2225	0.9997	163.1579	−18.4211	−144.7368
(400, 200, 100)	1012.6582	0.9955	0	0	60
(400, 100, 200)	1012.6582	0.9955	0	100	−40
(200, 100, 400)	932.7115	0.9955	168.4211	84.2105	−252.6316
(200, 400, 100)	1012.6582	0.9955	200	−200	60
(100, 200, 400)	932.7115	0.9955	268.4211	−15.7895	−252.6316
(100, 400, 200)	932.7115	0.9955	268.4211	−215.7895	−52.6316

Note: The negative values of (O_2-d_2) and (O_3-d_3) represent shortages of grade 2 and grade 3 products, respectively. These shortages are offset by excess products from higher-quality grades.

workstation, calculated as $\sum_{i=1}^{n}(\rho_i \times w_i)$, where ρ_i is the manufacturing cost per unit at workstation i.

(3) Inventory cost:

The inventory cost refers to the cost that arises when finished products of each grade remain in inventory after demand is satisfied and before they are sold. For Policy I, the number of excess products at grade g is given by $O_g - d_g$. In contrast, for Policy II, the excess amount at each grade is defined as the remaining amount after all downward substitution actions have been completed. Let τ_g be the inventory cost per unit per day for grade g, t_g the storage duration of grade g before being sold, and $r_{g,a}$ the number of grade g used to fulfil the demand of grade a, where $1 \le g < a \le h$. The inventory cost is calculated as $\sum_{g=1}^{h}((o_g - d_g - \sum_{a=g+1}^{h} r_{g,a}) \times \tau_g \cdot t_g)$.

(4) Revenue loss from downward substitution:

Since higher-quality grades possess greater value and are associated with higher market prices than lower-quality grades, using them to satisfy demand for lower-quality grades results in a revenue loss. Let p_g denote the market price of grade g, where $p_1 > p_2 > \ldots > p_h$. When a product of grade g is used to fulfil the demand of grade a ($g < a$), the revenue loss caused by downward substitution is defined as the price difference ($p_g - p_a$). The total revenue loss is calculated by summing the price differences over all downward substitution cases, shown in $\sum_{a=2}^{h}\sum_{g=1}^{a-1} r_{g,a}(p_g - p_a)$.

As mentioned, sales revenue is calculated by considering the revenue from satisfying current demand and the revenue obtained from selling excess products in future periods.Following the four cost components described above, the profit is $\sum_{g=1}^{h}(d_g \times p_g) + \sum_{g=1}^{h}((o_g - d_g - \sum_{a=g+1}^{h} r_{g,a}) \times p_g) - I \times MC - \sum_{g=1}^{h}((o_g - d_g - \sum_{a=g+1}^{h} r_{g,a}) \times$

$\tau_g t_g) - \sum_{a=2}^{h} \sum_{g=1}^{a-1} r_{ga}(p_g - p_a)$. The manufacturing cost per unit for all five workstations (ρ_1 to ρ_5) are 3.8, 5.5, 5.8, 7.6, and 12.4 USD, respectively; the material cost MC is 20.7 USD; the market price for grade 1 and grade 2, denoted by p_1 and p_2, are 120 and 60 USD, respectively; the holding cost is 1 USD per unit per day, and excess grade 1 and grade 2 are assumed to be sold after 30 days of storage. Taking $D = (550, 350)$ for example, for calculating the profit under Policy I, the costs are calculated as follows:

(1) material cost:

$$I \times MC = 1,615.1585 \times 20.7 = 33433.7809$$

(2) manufacturing cost:

$$\sum_{i=1}^{5} (\rho_i \times w_i) = 1,615.1585 \times 3.8 + 1,534.4006 \times 5.5 + 1,488.3685 \times 5.8 + 1,399.0665 \times 7.6 + 1,343.1038 \times 12.4 = 50,496.7354$$

(3) Inventory cost is calculated as follows:

$$\sum_{g=1}^{2} \left(\left(o_g - d_g - \sum_{a=g+1}^{2} r_{g,a} \right) \times \tau_g \cdot t_g \right) = ((797.6121 - 550 - 0) \times 1 \times 30) + ((350—350 - 0) \times 1 \times 30) = 7428.3630.$$

Then, sales revenue from the current order is equal to 87,000 USD ($550 \times 120 + 350 \times 60$); sales revenue for future demand is equal to 29,713.452 USD ($247.6121 \times 120 + 0 \times 60$). The profit is 25,354.5727 USD ($87,000 + 29,713.452 - 33,433.7809 - 50,496.7354 - 7,428.3630$).

For profit calculation under Policy II, the costs are first calculated as follows:

(1) material cost:

$$I \times MC = 1,266.6672 \times 20.7 = 26,220.0110$$

(2) manufacturing cost:

$$\sum_{i=1}^{5} (\rho_i \times w_i) = 1,266.6672 \times 3.8 + 1,203.3338 \times 5.5 + 1,167.2338 \times 5.8 + 1,097.1998 \times 7.6 + 1,053.3118 \times 12.4 = 39,601.4121$$

(3) Inventory cost is calculated as follows:

$$\sum_{g=1}^{2} \left(\left(o_g - d_g - \sum_{a=g+1}^{2} r_{g,\ a} \right) \times \tau_g \cdot t_g \right) = ((625.5170 - 550 - 75.5170) \times 1 \times 30) + (0 \times 1 \times 30) = 0$$

(4) Substitution cost is calculated as follows:

$$r_{12} \times (p_1 - p_2) = 75.5170 \times (120 - 60) = 4531.02$$

Then, sales revenue from the current order is equal to 87,000 USD ($550 \times 120 + 350 \times 60$). With no excess products, revenue from excess products is zero. Therefore, the profit is 16,647.5569 USD ($87,000 + 0 - 26,220.0110 - 39,601.4121 - 0 - 4531.02$).

Profit and system reliability results for the two policies and various demands for the two grades are summarised in Table 8. A trade-off between profit and system reliability is revealed: higher profit is obtained under Policy I with lower system reliability, whereas higher system reliability is obtained under Policy II with reduced profit. Furthermore, a set of experiments with $D = (550, 350)$ is conducted to investigate the effect of the storage duration of excess products before being sold, with the results summarised in Table 9. With increasing storage duration, Policy I yields lower profit than Policy II.

6.2. Managerial implications

There is a clear trade-off between system reliability and profit for Policy I and Policy II. Policy I is associated with higher profit but lower system reliability, whereas Policy II is associated with higher system reliability but lower profit. Based on the results for system reliability and profit, actionable insights for managers are summarised as follows:

1. Under a profit-oriented managerial objective, Policy I is a preferable option for managers. However, the main drawback of Policy I lies in its relatively low system reliability. To avoid an excessively low system reliability, a feasible approach is to reduce the demand for a specific grade. This reduction can be implemented at the order acceptance stage, where the manufacturer determines the maximum order quantity that can be accepted for each grade to achieve an acceptable level of system reliability, and then communicates the available supply quantity to customers. For example, if the

Table 8. Impact of different demands on profit and system reliability under two policies.

$D = (d_1, d_2)$	Policy I R_D	Policy II R_D	Policy I profit	Policy II profit	Profit difference (Policy I – Policy II)
(400, 500)	0	0.9486	24,649.3842	−1,352.4431	26,001.8273
(450, 450)	0	0.9486	24,884.4466	4,647.5569	20,236.8897
(500, 400)	0.0625	0.9486	25,119.5069	10,647.5569	14,471.9500
(500, 300)	0.8346	0.9655	22,529.6324	16,131.1614	6,398.4710
(400, 400)	0.0625	0.9655	21,999.5069	4,131.1614	17,868.3455
(550, 350)	0.6036	0.9486	25,354.5727	16,647.5569	8,707.0158
(600, 300)	0.8346	0.9486	25,589.6324	22,647.5569	2,942.0755
(650, 250)	0.9359	0.9359	25,659.0908	25,659.0908	0
(700, 200)	0.8346	0.8346	25,555.9417	25,555.9417	0
(750, 150)	0.6633	0.6633	25,452.7969	25,452.7969	0
(750, 337)	0.6633	0.6633	31,025.6879	30,393.2152	632.4727
(750, 338)	0.6036	0.6633	31,050.9808	30,338.3835	712.5973
(750, 339)	0.6036	0.6633	31,076.2858	30,283.5497	792.7361

Table 9. Profit comparison for different storage durations when $D = (550, 350)$.

Storage duration (days)	Profit of Policy I	Profit difference (Policy I–Policy II)
15	29,068.7542	12,421.1973
30	25,354.5727	8,707.0158
45	21,640.3912	4,992.8343
60	17,926.2097	1,278.6528
75	14,212.0281	−2,435.5288

Note: The profit of Policy II is $16,647.5569, as no excess products remain, regardless of the storage duration.

acceptable system reliability threshold is 0.8 when $D = (500, 400)$, system reliability under Policy I is below 0.1. By reducing the demand d_2 for grade 2 by 100 units, the system reliability increases by approximately 0.7, reaching 0.83 (See Table 4). In this case, the decision is made to decrease d_2 by 100 units, rather than reducing the demand d_1 for grade 1, because even if the demand d_1 for grade 1 is reduced to zero, the required input is still dominated by d_2, and the system reliability remains below 0.1.

2. Under a system reliability-oriented managerial objective, such as when fulfilling orders from important customers or those with urgent demands, where failing to meet demand could jeopardise customer satisfaction and long-term partnerships, ensuring system reliability becomes a priority in decision-making. In such cases, adopting Policy II is more appropriate, as it helps secure successful order fulfilment, even if it leads to lower profit. However, if prolonged storage reduces the profit under Policy I below that under Policy II, Policy II holds an absolute advantage. For example, when the storage duration increases from 30 days to 75 days, both system reliability and profit under Policy II outperform those under Policy I.

7. Conclusion

This study is mainly to assess the system reliability and profit of a manufacturing system in the context of fulfilling demands for products of different grades under the same-grade fulfilment (Policy I) and downward substitution (Policy II). System reliability refers to the probability that the manufacturing system can meet the demands of all grades, serving as an intuitive performance index for order acceptance decisions. To evaluate system reliability, the manufacturing system is modelled as a multistate flow network (MFN). Next, we derive the lower capacity vector (LCV) based on the relationship between the workload and the multiple capacities of each workstation, enabling efficient calculation of system reliability. Experimental results reveal a clear trade-off between system reliability and profit. Policy II improves system reliability and reduces inventory, but results in lower profit. Conversely, Policy I achieves higher profit despite lower system reliability; nevertheless, prolonged storage duration may gradually offset its profit advantage.

The main contributions of this study are threefold. First, the gap in current system reliability assessment studies is bridged, as production grading and demand fulfilment policies have not been addressed, and studies on product grading and downward substitution have overlooked the multistate nature of manufacturing systems. Therefore, a novel MFN model is established to consider the fulfilment of multi-grade orders. Second, an algorithm is developed to generate LCV for computing system reliability under Policy I (same-grade fulfilment) and Policy II (downward substitution), with a proof provided for deriving the amount of input required to fulfil demands under Policy II. Finally, the practicality of the proposed algorithm is demonstrated through a case study of a ball screw pair manufacturing system. By jointly considering system reliability and profit, managers can identify the most appropriate policy. When profit is prioritised, Policy I is preferred, whereas Policy II is adopted to ensure higher system reliability and secure order fulfilment.

Furthermore, based on the findings and assumptions of this study, several directions for future

research are proposed. First, since there is a trade-off between system reliability and profit under the two policies, manufacturers are interested in identifying the optimal policy, which could be Policy I, Policy II, or a partial downward substitution policy representing varying levels of downward substitution, by balancing these two conflicting objectives. Consequently, the current problem can be extended to a bi-objective optimisation problem that simultaneously maximises profit and system reliability. Second, the impact of rework on products with multiple grades requires further exploration, especially in terms of which product grades are suitable for rework, the potential improvement to higher-quality grades, the risk of becoming scrap, and the associated costs. Finally, the current assumption is that all KQCs are statistically independent. However, in reality, KQCs may not be statistically independent, and it is worth exploring the impact of this aspect on the calculations of the percentage for each grade.

Author contributions

CRediT: **Hsuan-Yu Chen**: Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft; **Yi-Kuei Lin**: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

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Appendix A: Proof of Proposition 1

Proposition 1. If $\frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i} > \frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}$, then $\frac{d_1+d_2}{(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i} > \frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}$.

Proof:

$$\frac{d_1+d_2}{(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i}-\frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}=\frac{Q_1\times\prod_{i=1}^{n}\theta_i\times(d_1+d_2)-d_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{Q_1\times\prod_{i=1}^{n}\theta_i\times d_1+Q_1\times\prod_{i=1}^{n}\theta_i\times d_2-Q_1\times\prod_{i=1}^{n}\theta_i\times d_1-d_1\times Q_2\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{Q_1\times\prod_{i=1}^{n}\theta_i\times d_2-d_1\times Q_2\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{\left(\frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i}-\frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}\right)\times Q_1\times\prod_{i=1}^{n}\theta_i\times Q_2\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

This value is positive, when $\frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i} > \frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}$.

Appendix B: Proof of Proposition 2

Proposition 2. If $\frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i} < \frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i}$, then $\frac{d_1+d_2}{(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i} > \frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i}$

Proof:

$$\frac{d_1+d_2}{(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i}-\frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i}=\frac{Q_2\times\prod_{i=1}^{n}\theta_i\times(d_1+d_2)-d_2(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{d_1\times Q_2\times\prod_{i=1}^{n}\theta_i+d_2\times Q_2\times\prod_{i=1}^{n}\theta_i-d_2\times Q_1\times\prod_{i=1}^{n}\theta_i-d_2\times Q_2\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{d_1\times Q_2\times\prod_{i=1}^{n}\theta_i-d_2\times Q_1\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

$$=\frac{\left(\frac{d_1}{Q_2\times\prod_{i=1}^{n}\theta_i}-\frac{d_2}{Q_1\times\prod_{i=1}^{n}\theta_i}\right)\times Q_1\times\prod_{i=1}^{n}\theta_i\times Q_2\times\prod_{i=1}^{n}\theta_i}{Q_1(Q_1+Q_2)\times\prod_{i=1}^{n}\theta_i\times\prod_{i=1}^{n}\theta_i}$$

This value will be positive, when $\frac{d_1}{Q_1\times\prod_{i=1}^{n}\theta_i} > \frac{d_2}{Q_2\times\prod_{i=1}^{n}\theta_i}$.

Appendix C: Acronyms and notations

Acronyms

KQC	key quality characteristic
MFN	multistate flow network
LCV	lower capacity vector
LSL	lower specification limit
USL	upper specification limit
WIP	work in progress

Notations

β	number of KQCs of the product
h	number of tolerances of a specific KQC specified by all orders.
$\underline{s}_g^{\alpha}$	minimum specification of the gth tolerance of the αth KQC, $\alpha = 1, 2, \ldots, \beta$, $g = 1, 2, \ldots, h$.
$\bar{s}_g^{\alpha}$	maximum specification of the gth tolerance of the αth KQC, $\alpha = 1, 2, \ldots, \beta$, $g = 1, 2, \ldots, h$.
S	$\{[\underline{s}_g^{\alpha}, \bar{s}_g^{\alpha}] \mid \alpha = 1, 2, \ldots, \beta, g = 1, 2, \ldots, h\}$: set of all tolerances for all KQCs.
X_{α}	a random variable represents the specifications of the αth KQC, $\alpha = 1, 2, \ldots, \beta$.
$\hat{\mu}_{X_{\alpha}}$	the estimated mean of the αth KQC, $\alpha = 1, 2, \ldots, \beta$.
$\hat{\sigma}_{X_{\alpha}}$	the estimated standard deviation of the αth KQC, $\alpha = 1, 2, \ldots, \beta$.
q_g^{α}	the probability of the αth KQC meets the gth tolerance, $\alpha = 1, 2, \ldots, \beta$, $g = 1, 2, \ldots, h$.
Q_g	the percentage of products belong to grade g in produced products, $g = 1, 2, \ldots, h$.
$\Phi(\cdot)$	cumulative distribution function of standard normal distribution.
N	set of nodes(buffers).
n	number of workstations in the manufacturing system
e_i	ith arc(workstation), $i = 1, 2, \ldots, n$.
E	$\{e_i \mid i = 1, 2, \ldots, n\}$: set of edges (workstation).
G	(**N**, **E**, **S**) an MFN.
θ_i	yield rate of the ith arc(workstation), $i = 1, 2, \ldots, n$.
O_q	the amount of grade q products produced by the manufacturing system, $q = 1, 2, \ldots, s$.
d_g	the demand for grade g products, $g = 1, 2, \ldots, h$.
D	(d1, d2, … , dh): demands for products of all grades from all orders.
I	the amounts of raw materials input into the manufacturing system.
I_g	the amount of raw material input that guarantees the satisfaction of the demand of grade g through only grade g products, $g = 1, 2, \ldots, h$.
I'_{η}	the amount of input that guarantees the satisfaction of the demand for grade η through downward substitution, $\eta = 2, 3, \ldots, h$.
m_i	number of identical machines in the ith workstation, $i = 1, 2, \ldots, n$.
t_i	number of functioning machines in workstation i, $i = 1, 2, \ldots, n$.
π_i	machine availability at the ith workstation, $i = 1, 2, \ldots, n$.
w_i	workload of the ith workstation i, $i = 1, 2, \ldots, n$.
z_i	current capacity of e_i, $i = 1, 2, \ldots, n$.
Z	$(z_1, z_2, \ldots, z_n)$: the capacity vector.
$O_g(Z)$	the maximal cumulative output of products from grade 1 through grade r products when the amount of raw material input is equal to I, given the capacity vector Z, $r = 1, 2, \ldots, s$.
Ω_{I}	set of capacity vectors that provide sufficient capacity to satisfy D under Policy I.
Ω_{II}	set of capacity vectors that provide sufficient capacity to satisfy D under Policy II.
$\underline{z}_i$	the minimal capacity that the ith workstation can provide to satisfy the workload w_i.
$\underline{Z}$	$(\underline{z}_1, \underline{z}_2, \ldots, \underline{z}_n)$: an LCV in Ω_{I} (under Policy I) or Ω_{II} (Policy II).
R_D	system reliability, i.e., the probability that the MFN can meet the demand across all grades D.
MC	per-unit cost of a raw material
ρ_i	manufacturing cost per unit at workstation i, $i = 1, 2, \ldots, n$.
τ_g	inventory cost per unit per day for grade g, $g = 1, 2, \ldots, h$.
t_g	storage duration of grade g before being sold, $g = 1, 2, \ldots, h$.
$r_{g,a}$	the amount of grade g used to fulfil the demand of grade a, where $1 \le g < a < h$.
p_g	the market price of grade g, $g = 1, 2, \ldots, h$.

Appendix D: System reliability assessment details under Policy I

INPUT: (**N**, **E**, **S**), $D = (550, 350)$

Step 1. The calculation of the percentage of each KQC meeting all different tolerances is as follows:

$$q_1^1 = \Phi\left(\frac{0.0008 - 0.0007}{0.0001}\right) - \Phi\left(\frac{0 - 0.0007}{0.0001}\right) = 0.8413,$$

$$q_2^1 = \Phi\left(\frac{0.0009 - 0.0007}{0.0001}\right) - \Phi\left(\frac{0 - 0.0007}{0.0001}\right) = 0.9999,$$

$$q_1^2 = \Phi\left(\frac{36.2 - 36.124}{0.052}\right) - \Phi\left(\frac{36 - 36.124}{0.052}\right) = 0.9195,$$

$$q_2^2 = \Phi\left(\frac{36.22 - 36.124}{0.052}\right) - \Phi\left(\frac{35.98 - 36.124}{0.052}\right) = 0.9647,$$

$$q_1^3 = \Phi\left(\frac{27.05 - 26.86}{0.04}\right) - \Phi\left(\frac{26.8 - 26.86}{0.04}\right) = 0.9332,$$

$$q_2^3 = \Phi\left(\frac{27.06 - 26.86}{0.04}\right) - \Phi\left(\frac{26.78 - 26.86}{0.04}\right) = 0.9772,$$

$$q_1^4 = \Phi\left(\frac{40.51 - 40.481}{0.02}\right) - \Phi\left(\frac{40.45 - 40.481}{0.02}\right) = 0.8659,$$

$$q_2^4 = \Phi\left(\frac{40.52 - 40.481}{0.02}\right) - \Phi\left(\frac{40.44 - 40.481}{0.02}\right) = 0.9542.$$

Step 2. The percentages of the two grades are

$$Q_1 = \prod_{\alpha=1}^{4} q_1^{\alpha} = 0.6251,$$

$$Q_2 = \prod_{\alpha=1}^{4} q_2^{\alpha} - \prod_{\alpha=1}^{4} q_1^{\alpha} = 0.8994 - 0.6251 = 0.2743.$$

Step 3. The amount of input I fulfilling $D = (550, 350)$ is

$$I = \max\left\{\frac{550}{0.6251 \times 0.79}, \frac{350}{0.2743 \times 0.79}\right\} = 1615.1585.$$

The workload w_1 of the first workstation is 1615.1585 units.

Step 4. The workloads for the rest of the workstations are listed as follows:

$$w_2 = 1615.1585 \times 0.95 = 1534.4006,$$
$$w_3 = 1534.4006 \times 0.97 = 1488.3686,$$
$$w_4 = 1488.3686 \times 0.94 = 1399.0665,$$
$$w_5 = 1399.0665 \times 0.96 = 1343.1038.$$

Step 5. All workloads are below the corresponding maximal capacities listed in Table 2, i.e., $w_i < m_i$, $i = 1, 2, 3, 4, 5$.

Step 6. The required capacity at the first workstation to process the workload is 1800 ($\underline{z}_1 = 1800 \ge 1615.1585 > 1500$), and those at the remaining workstations are

determined similarly. Consequently, the LCV is obtained as $\underline{Z}=(1800, 1750, 1500, 1575, 1500)$.

OUTPUT: $\underline{Z}=(1800, 1750, 1500, 1575, 1500)$.

System reliability is assessed *via*

$$\begin{aligned} R_{D=(550,\ 350)} &= \Pr\{Z|Z \geq \underline{Z}\} \\ &= \Pr\{Z|(z_1,\ z_2,\ \ldots,\ z_n) \\ &\quad \geq (1800,\ 1750,\ 1500,\ 1575,\ 1500)\} \\ &= 0.8745 \times 0.9227 \times 0.8745 \times 0.8965 \times 0.9541 \\ &= 0.6036. \end{aligned}$$

Appendix E: System reliability assessment details under Policy II

The system reliability of the MFN under Policy II, given demands of 550 and 350 units for the two grades, is obtained as follows:

INPUT: (**N**, **E, S**), $D = (550, 350)$

Step 1. The percentages of each KQC meeting all different tolerances are the same as the results presented in Appendix D, Step 1.

Step 2. The percentages of two grades are the same as the result shown in Appendix D.

Step 3. The amount of input I fulfilling $D = (550, 350)$ is

$$\begin{aligned} I &= \max\left\{\frac{550}{0.6251 \times 0.79},\ \frac{550+350}{(0.6251+0.2743) \times 0.79}\right\} \\ &= 1266.6672. \end{aligned}$$

The workload w_1 of the first workstation is 1266.6672.

Step 4. The workloads for the subsequent workstations are as follows:

$w_2 = 1203.3338$,
$w_3 = 1167.2338$,
$w_4 = 1097.1998$,
$w_5 = 1053.3118$.

Step 5. All workloads are below the corresponding maximal capacities listed in Table 2, i.e., $w_i < m_i$, $i = 1, 2, 3, 4, 5$.

Step 6. The required capacity at the first workstation to process the workload is 1500 ($\underline{z}_1 = 1500 \geq 1266.6672 > 1200$), and those at the remaining workstations are determined similarly. Therefore, the LCV is obtained as $\underline{Z}=(1500, 1400, 1250, 1125, 1200)$.

OUPUT: $\underline{Z}=(1500, 1400, 1250, 1125, 1200)$

Then, system reliability is assessed *via*

$$\begin{aligned} R_{D=(550,\ 350)} &= \Pr\{Z|Z \geq \underline{Z}\} \\ &= \Pr\{Z|\ (z_1,\ z_2,\ \ldots,\ z_n) \\ &\quad \geq\ (1500,\ 1400,\ 1250,\ 1125,\ 1200)\} \\ &= 0.9486. \end{aligned}$$